



Advanced Certification Programme in Data Science Business Analytics



Week 8

Visualisation in Power BI



Topics Covered

- Visuals in Power BI
- Custom Visuals
- Formatting Visuals
- Q and A

Async Recap

1. Power BI simplifies data analysis and reporting

Power BI Desktop helps turn raw data into meaningful insights. It enables easy data import, transformation, and visualisation—empowering users to make data-driven decisions.

2. Power Query enables efficient data preparation

With Power Query, you can clean, transform, and shape data before analysis. Remove errors, handle missing values, and preview changes to ensure data quality.

3. Data modeling ensures accurate insights

Use Power Pivot to create structured data models and relationships between tables. This step is essential for producing accurate calculations and visualisations.

4. Visualisations make data intuitive and engaging

Bar charts, line graphs, pie charts, and maps help present data clearly. Customisation options like tooltips, formatting, and slicers improve user experience and understanding.

5. Dashboards and interactivity enhance user experience

Combine visuals into dashboards, use slicers for filtering, and enable cross-filtering for deeper exploration. Always test your visuals to ensure clarity and functionality.

The Power of Data Visualisation

Transforming Information into Insights



- Represents information and data visually
- Uses charts, graphs and maps for better insights
- Helps identify trends, outliers and patterns
- Enhances data interaction and understanding

The Need for Visualisation

Making Data Understandable and Actionable

Simplifies complex data

- Breaks down large amounts of raw data into easy-to-understand visuals
- Uses charts and graphs to make complex information clearer
- Helps quickly spot trends and patterns that may be hard to see in spreadsheets

Better decision-making

- Provides quick insights to help decision-makers understand data faster
- Enables interactive exploration by clicking on different parts of visuals
- Leads to better and quicker choices with a deeper understanding of details

The Need for Visualisation

Enhancing Communication and Exploration

Tells a story

- Communicates key points clearly and engagingly, making it easier to explain findings
- Transforms numbers into visuals, making presentations more interesting and simpler to follow

Speeds up exploration

- Provides real-time insights, allowing users to track live data updates for sales and performance
- Enables interactive filtering and adjustments, helping to quickly find specific answers and trends

The Need for Visualisation

Making Data Clear, Actionable and Collaborative

Highlights important metrics

- Monitors progress on sales targets, project milestones and key business metrics
- Spot trends by comparing current results against targets or past performance for better insights

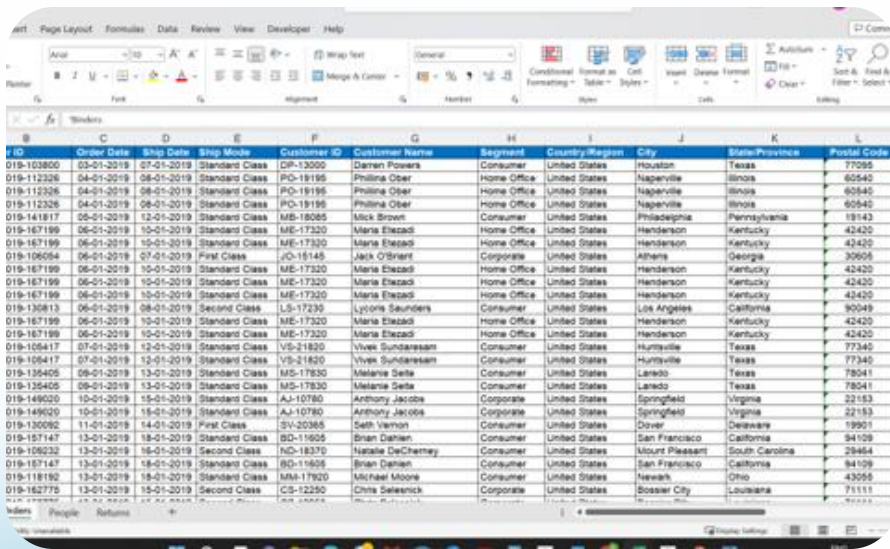
Encourages data-driven thinking

- Shares reports and dashboards effortlessly, ensuring teams work with the same reliable data
- Enables all users to create and analyse reports without needing technical expertise

The Usage of Visualisation

Transforming Raw Data into Actionable Insights

Raw data



ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State/Province	Postal Code
019-103800	03-01-2019	07-01-2019	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston	Texas	77060
019-112328	04-01-2019	08-01-2019	Standard Class	PO-19195	Phyllina Ober	Home Office	United States	Naperville	Illinois	60540
019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	Phyllina Ober	Home Office	United States	Naperville	Illinois	60540
019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	Phyllina Ober	Home Office	United States	Naperville	Illinois	60540
019-141917	06-01-2019	12-01-2019	Standard Class	MB-18055	Mike Brown	Consumer	United States	Philadelphia	Pennsylvania	19143
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-106054	06-01-2019	07-01-2019	First Class	JO-16145	Jack O'Brian	Corporate	United States	Atlanta	Georgia	30605
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-130813	06-01-2019	08-01-2019	Second Class	LS-17235	Lycoris Saunders	Consumer	United States	Los Angeles	California	90049
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	Maria Eliazad	Home Office	United States	Henderson	Kentucky	42420
019-108417	07-01-2019	12-01-2019	Standard Class	VS-21820	Vivek Sundaresan	Consumer	United States	Huntsville	Texas	77340
019-108417	07-01-2019	12-01-2019	Standard Class	VS-21820	Vivek Sundaresan	Consumer	United States	Huntsville	Texas	77340
019-136405	09-01-2019	13-01-2019	Standard Class	MS-17630	Melanie Sette	Consumer	United States	Laredo	Texas	78041
019-136405	09-01-2019	13-01-2019	Standard Class	MS-17630	Melanie Sette	Consumer	United States	Laredo	Texas	78041
019-149020	10-01-2019	15-01-2019	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield	Virginia	22153
019-149020	10-01-2019	15-01-2019	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield	Virginia	22153
019-130692	11-01-2019	14-01-2019	Standard Class	SV-20365	Seetha Vernon	Consumer	United States	Ozark	Delaware	19901
019-157147	13-01-2019	18-01-2019	Standard Class	BD-11605	Brian Dahan	Consumer	United States	San Francisco	California	94109
019-108232	13-01-2019	16-01-2019	Second Class	ND-18370	Natalie DeCherney	Consumer	United States	Mount Pleasant	South Carolina	29664
019-187147	13-01-2019	18-01-2019	Standard Class	BD-11605	Brian Dahan	Consumer	United States	San Francisco	California	94109
019-118192	13-01-2019	18-01-2019	Standard Class	MM-17920	Michael Moore	Consumer	United States	Newark	Ohio	43055
019-162775	13-01-2019	18-01-2019	Second Class	CS-12255	Chris Selenick	Corporate	United States	Bossier City	Louisiana	71111

Same data with visualisation

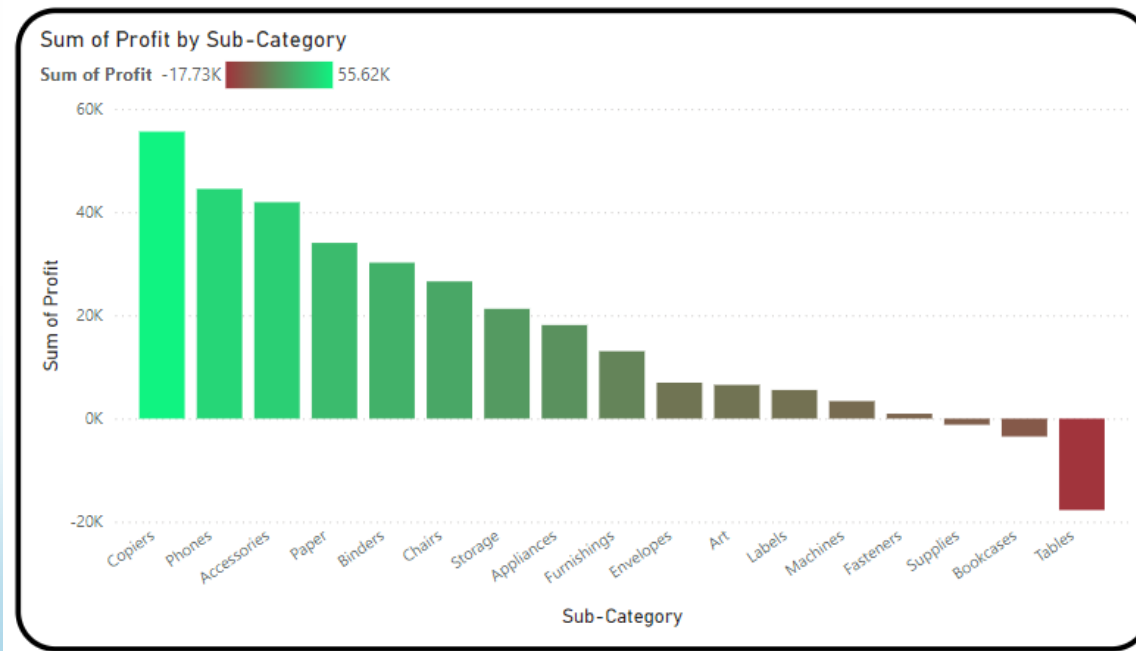


- Visualisations simplify complex data and make it easier to understand
- Help users identify patterns, trends and insights at a glance
- Enable faster, more informed decision-making
- Improve collaboration by making data accessible to everyone

Visuals in Power BI

Column Chart in Power BI

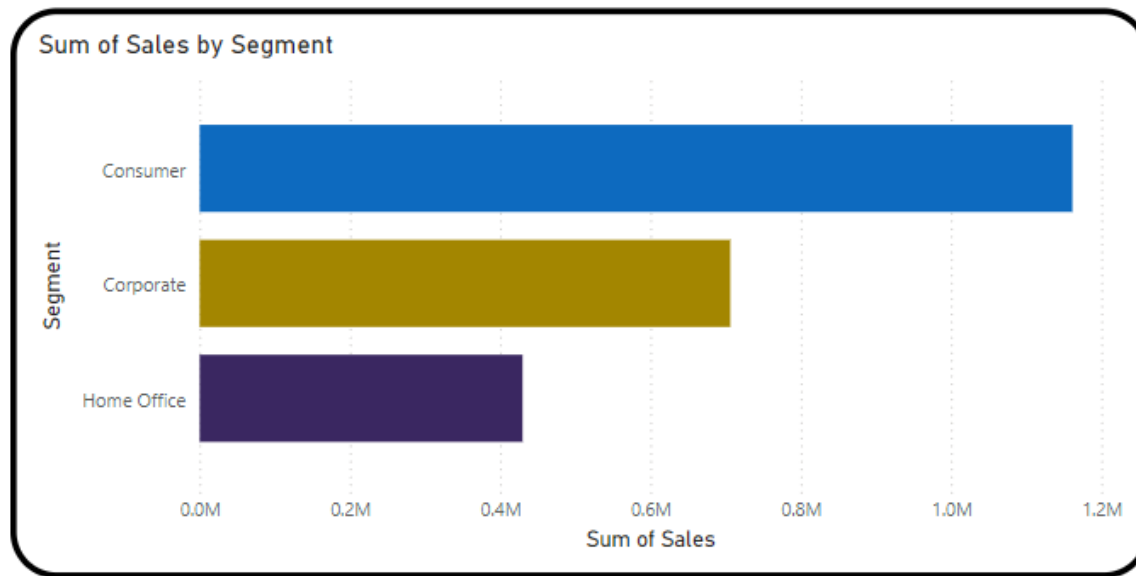
Visualising Data Comparisons



- Compare data across categories or time periods
- Show differences between groups
- Analyse sales by product or revenue by region
- Represent each category with a separate column

Bar Chart in Power BI

Emphasising Category Comparisons



- Compare values across different categories with ease
- Use horizontal bars to visually represent data effectively
- Highlight key differences for better analysis
- Work well when category labels are long or detailed

Card Visual in Power BI

Tracking Key Metrics at a Glance

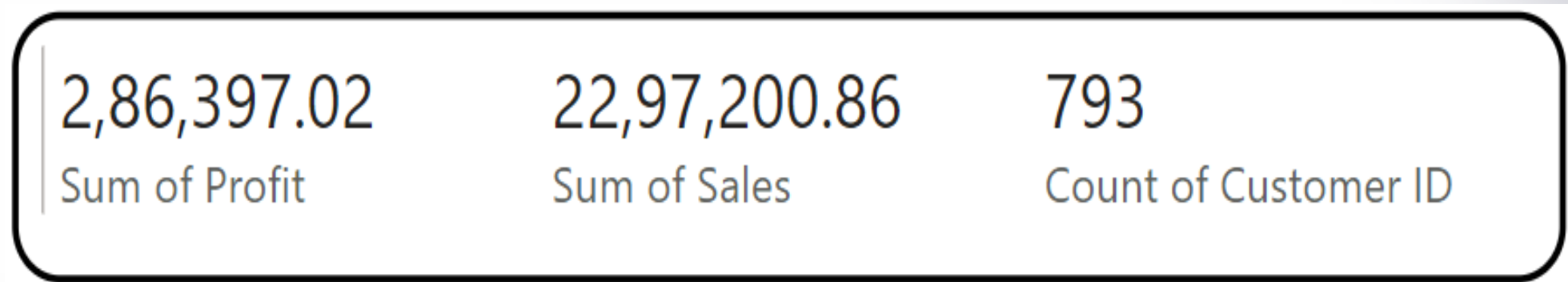
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Sum of Profit

- Display a single fact or data point clearly
- Highlight important metrics like total sales or market share
- Use for quick insights in dashboards and reports
- Focus on essential numbers for better decision-making

Multi-Row Card

Display Multiple Data Points Per Row



- Display one or more data points in a row
- Work similarly to card visual with additional values
- Present multiple metrics in a compact format
- Enhance data representation in Power BI

Demo – Visualisation

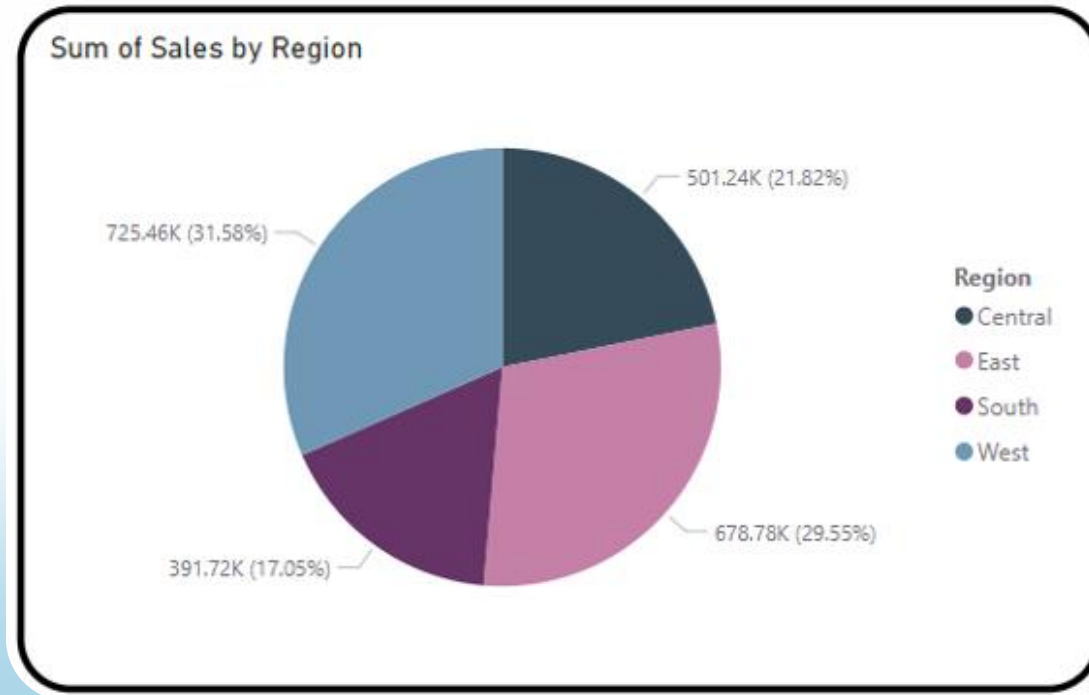
Understanding Charts and Cards in Power BI



- Create column chart, bar chart, card visual and multi-row card visual
- Demonstrate their use cases and applications
- Showcase data insights effectively

Understanding Pie Charts

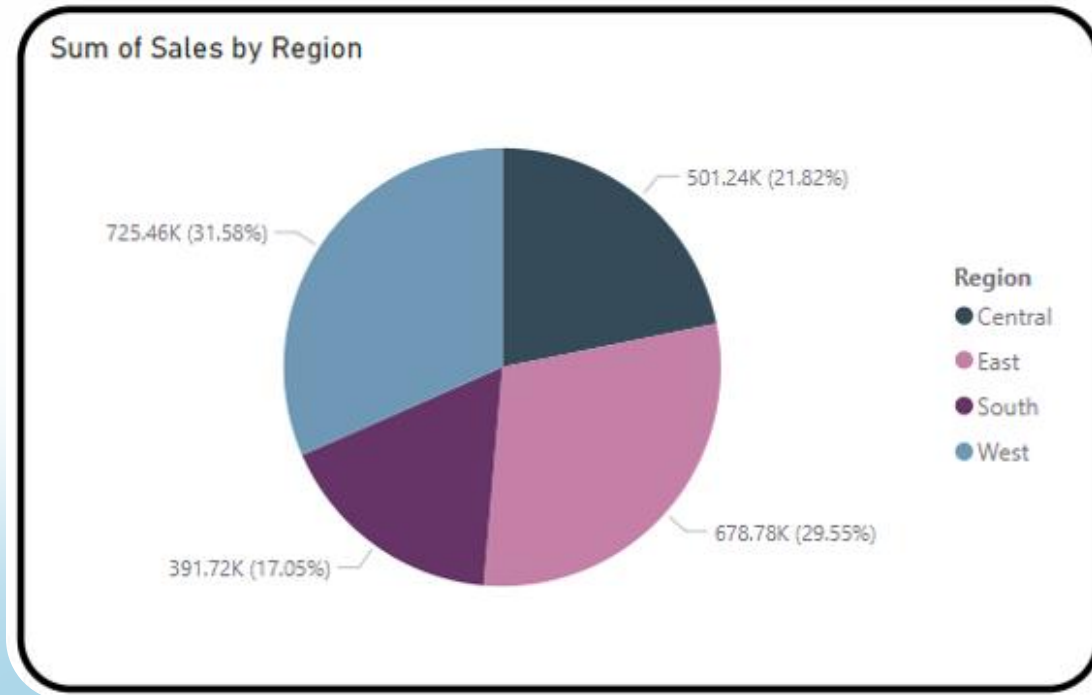
Proportional Data Visualisation in Power BI



- Show the proportional distribution of a whole in an easy-to-read format
- Compare the contribution of each category within a dataset
- Use for part-to-whole relationships such as market share or sales distribution
- Visualise simple data insights at a glance

Understanding Doughnut Charts

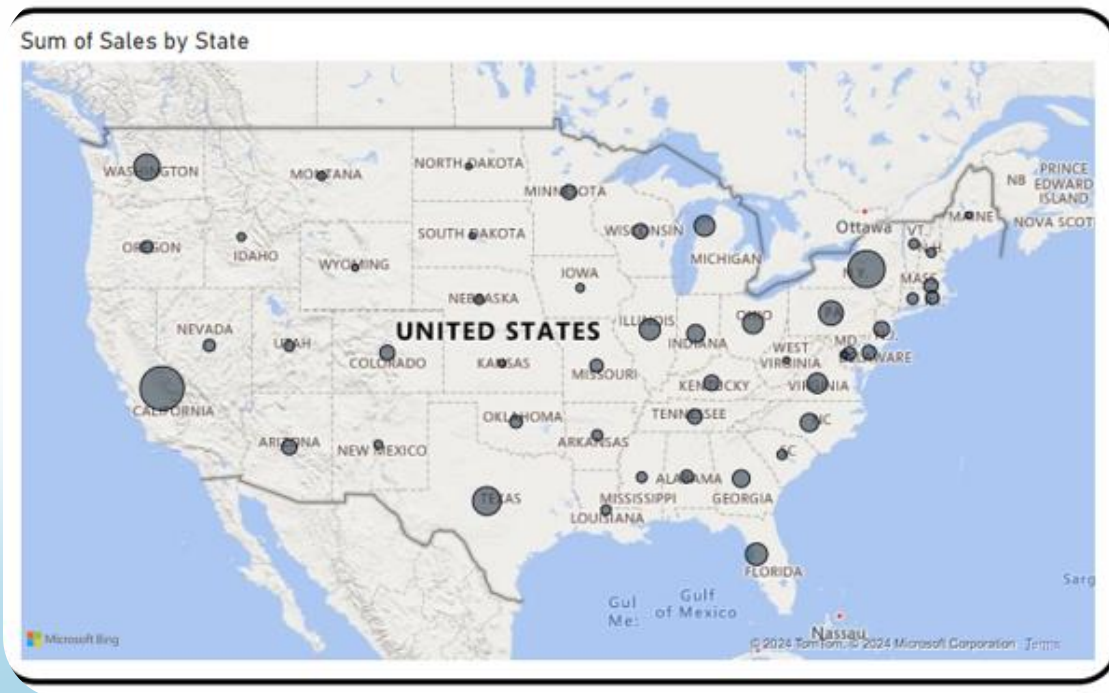
A Variation of Pie Charts for Better Visuals



- Show the relationship of parts to a whole in a circular format
- Keep the centre blank to add labels or icons for better readability
- Use for visualising proportions with a cleaner and modern look
- Compare multiple categories efficiently in a compact space

Geographic Data Representation

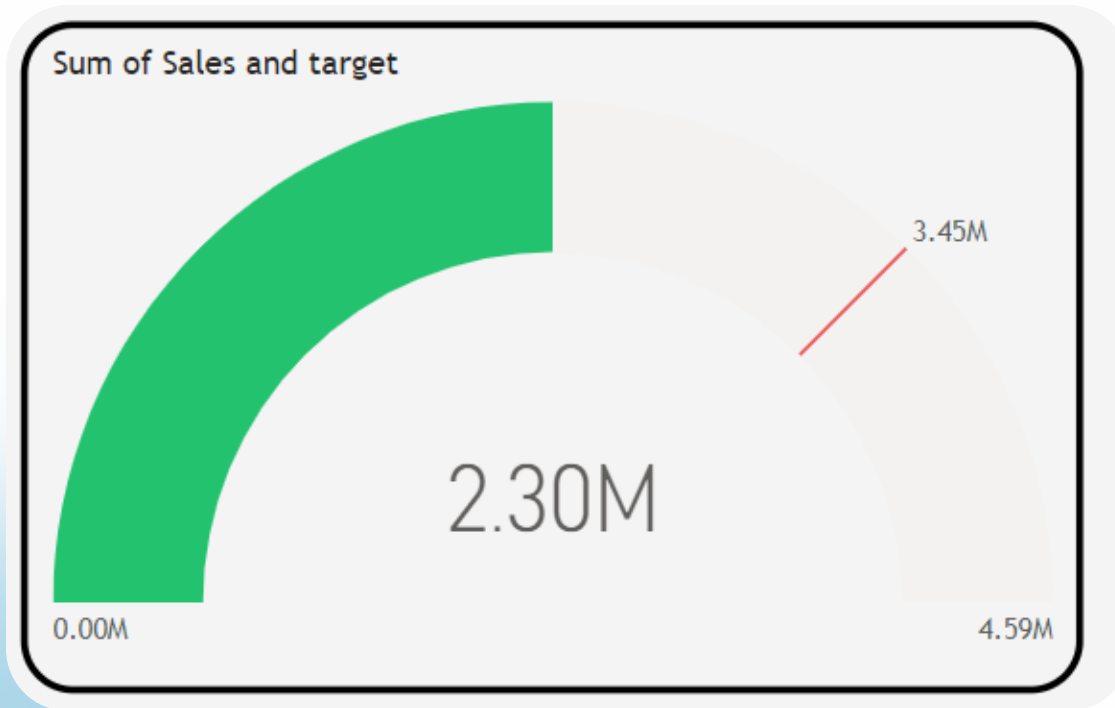
Visualising Information with Maps



- Use a basic map to link categorical and numerical data
- Display information with spatial locations
- Visualise data distribution across regions effectively

Gauge Chart for Performance Tracking

Track Progress by Comparing Target vs Actual Values



- Compare the difference between target values and actual values
- Track the progress made towards achieving specific goals effectively
- Identify how much work has been completed and how far you are from reaching the target

Data Visualisation Demo

Explore Different Chart Types and Their Applications

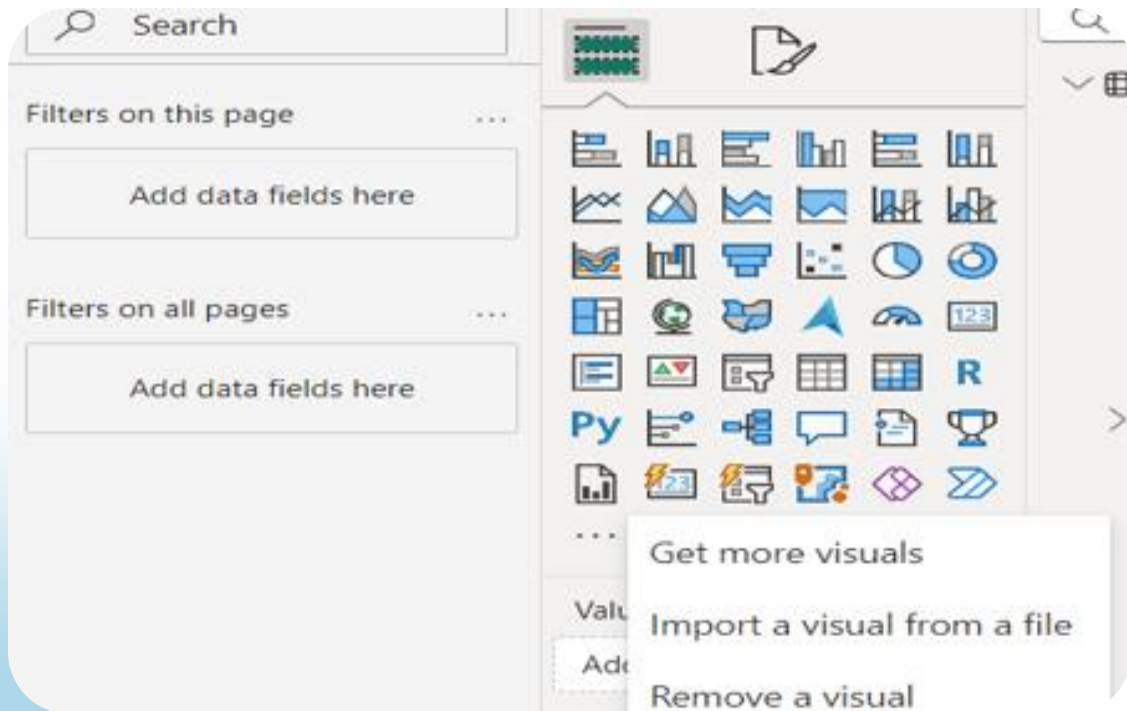


- Demonstrates pie, doughnut, map and gauge Charts
- Showcases their use cases and applications
- Helps in understanding data representation visually
- Enhances decision-making with effective visuals

Custom Visuals

Custom Visuals in Power BI

Expanding Visualisation Capabilities



- Power BI supports custom visuals beyond built-in options
- Download visuals from the office Store and integrate them
- Community and Microsoft contribute to these visuals
- Enhances dashboards with tailored data representation

Expanding Visual Options in Power BI

Import Custom Visuals for Better Insights

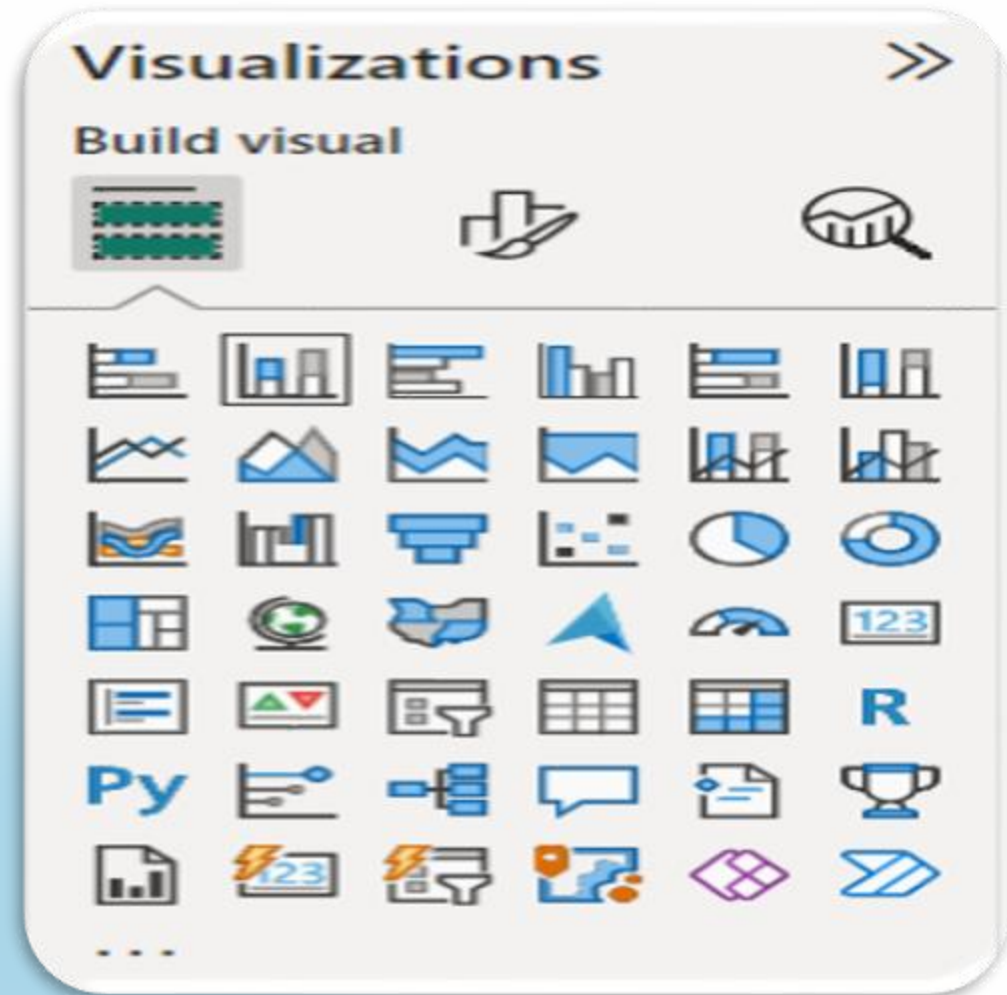


- Download custom visuals from the office Store
- Use them in your Power BI dashboards
- Created by Microsoft and the Power BI community
- Click on a visual, then select 'Add' to use it

Formatting Visuals

Formatting Visuals in Power BI

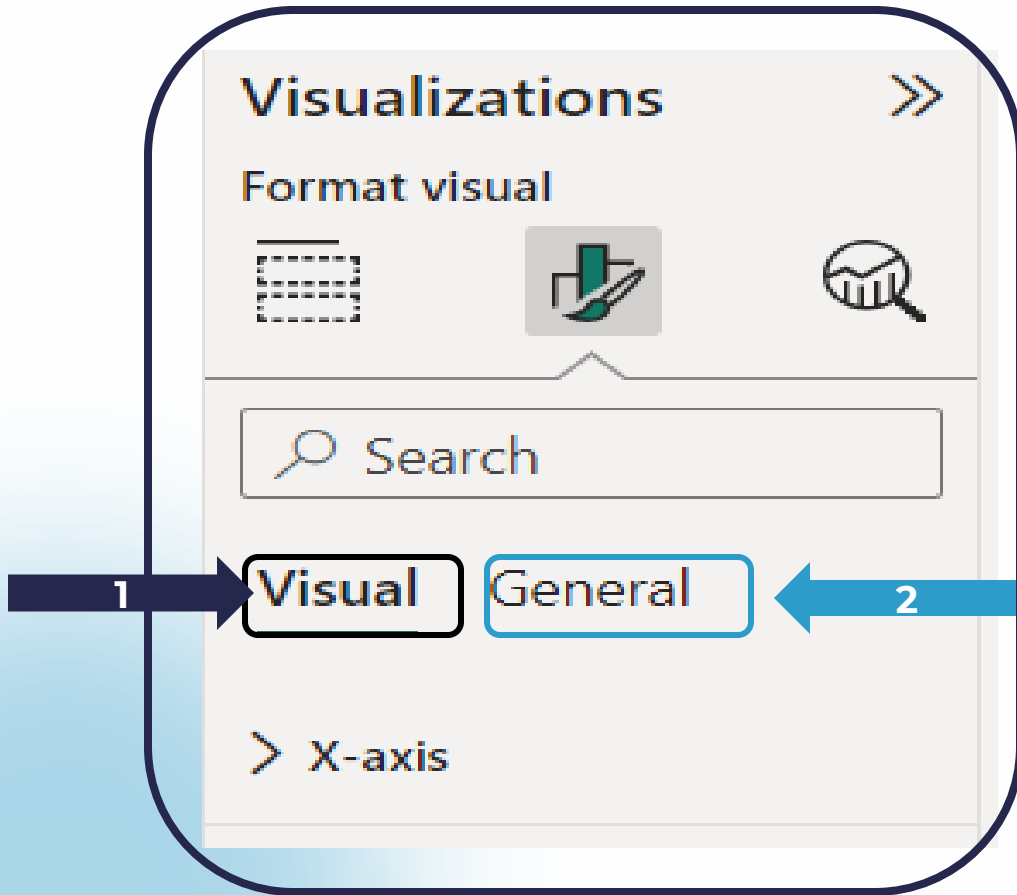
Customising Charts for Better Insights



- Adjust look, feel and behaviour of visuals
- Improve clarity and highlight key insights
- Align with brand or design preferences
- Use the Format Visual tab under the Visualisation Pane

Formatting Options in Power BI

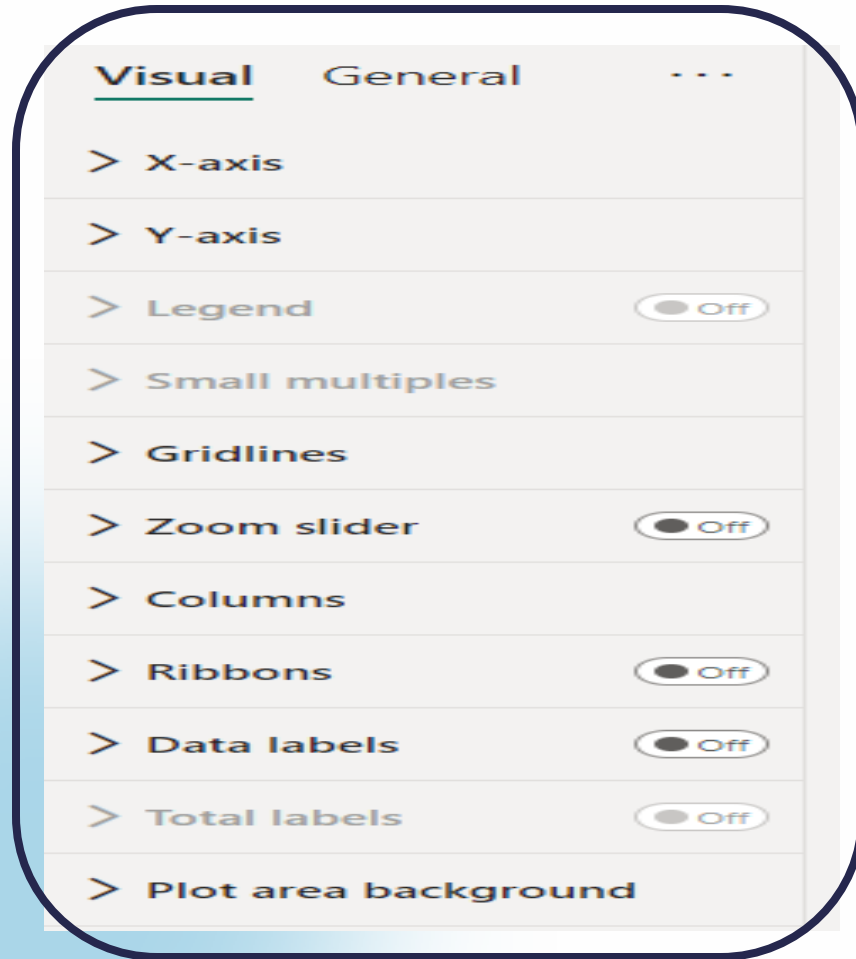
Customise Your Visuals with Ease



- Select a visual and go to "Format your visual"
- Two main categories:
- **Visual:** Adjust appearance and design
- **General:** Modify layout and position

Enhance Visuals with Formatting

Personalise Colours, Labels and Legends



- Adjust axes, gridlines and titles
- Modify labels, legends and colours
- Enable or disable zoom sliders
- Format columns, ribbons and data labels
- Enhance readability with plot area settings

Enhance Visual Adjustments

Customise Background Borders and Effects



- Modify background, borders & shadows
- Adjust size & position of visuals
- Customise title, properties & effects
- Enable tooltips & header icons
- Reset formatting to default

Visual Formatting Demo

Learn to Customise Visuals with Ease



- Format visuals for better clarity
- Use Format Painter for quick styling
- Multi-row cards display multiple data points per row
- Compare normal vs multi-row card visuals

Q & A

Thank you